Basics 69: Finite-Volume Methods

IX - Computational physics methods

Planned textbook-style prerequisite note for the Kinetic MHD-PINN computational-plasma-physics curriculum

Curriculum scaffold - not yet a completed textbook chapter. This file reserves the scope, learning sequence, and advanced-module connections for Basics 69. The completed version will begin from first principles, derive its central equations, include physical intuition and worked examples, and connect each derivation to numerical implementation. No placeholder statement in this scaffold should be cited as a finished derivation.

Contents

1 Role in the curriculum

This note belongs to **IX - Computational physics methods**. Its purpose is to make the later simulation modules readable to a student who has no prior plasma-physics background. The principal downstream connections are: **All numerical modules, especially Modules 3, 4, 6, 8, and 9**.

2 Learning objectives

After the completed note is written and studied, a reader should be able to:

- Integral conservation.
- Control volumes.
- Cell-centered variables.
- Face fluxes.
- Conservative discretization.
- Boundary fluxes.
- Module 4 radial operator.
- Explain which assumptions are physical approximations and which choices are purely numerical.
- Reproduce the main derivations and verify dimensional consistency.
- Identify where the concepts enter the Python and MATLAB implementations.

3 Planned textbook chapter structure

1. **Chapter 1: Integral conservation.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.
2. **Chapter 2: Control volumes.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.
3. **Chapter 3: Cell-centered variables.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.
4. **Chapter 4: Face fluxes.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.
5. **Chapter 5: Conservative discretization.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.
6. **Chapter 6: Boundary fluxes.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.
7. **Chapter 7: Module 4 radial operator.** Definitions, physical interpretation, derivations, worked examples, and computational implications will be developed.

8. **Worked examples and limiting cases.** Analytic checks, order-of-magnitude estimates, and common misconceptions.
9. **Computational laboratory.** A language-neutral numerical exercise with parallel Python and MATLAB implementations where appropriate.
10. **Summary, symbol table, and exercises.** Conceptual questions, derivations, coding exercises, and verification tasks.

4 Expected derivation standard

The completed note will not merely quote final equations. It will state the starting assumptions, define every symbol, show intermediate algebra, identify boundary or initial conditions, test units and limiting cases, and distinguish exact identities from reduced-model closures. Numerical formulas will be derived from the continuous equations before code is introduced.

5 Repository integration

The planned source and PDF names are:

```
basics_69_finite_volume_methods.tex
basics_69_finite_volume_methods.pdf
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Advanced module notes may list this document in a prerequisite table. Once the note is completed, those references should become clickable PDF links and should state exactly which sections are required rather than treating the entire note as mandatory.

6 Completion checklist

This scaffold will be considered complete only when it contains:

- a detailed symbols-and-acronyms table;
- first-principles derivations of the key equations;
- physical interpretation and order-of-magnitude examples;
- at least one fully worked analytic example;
- at least one computational exercise or reproducible notebook/script;
- verification questions and common-error warnings;

- references to authoritative textbooks, papers, or official software documentation.